



Inferential Statistics

Making Good Inferences from Good Stats

Colin Hannigan

CSB/SJU

POLS 222: Slide Deck #14

The questions we ask determine the answers we get.

“Good” survey/poll questions:

1. Are easily understood
2. Are administered consistently
3. Clarify the type of answer needed
4. Avoid over-burdening respondents
5. Rely on information that respondents have access to
6. Maximize respondent willingness to answer

Some questions are intentionally misleading (e.g., push polls about that damn Dr. Kevin Simpson...)

The Pitfalls of Survey/Polling Research

Satisficing: recalling just enough to provide an answer to the question to get through the survey quickly. (e.g. but what about inaccurate observations?)

Social desirability: choosing answers that present one in a favorable way, by talking up the good things, and minimizing the bad things.

Respondent error: People lie, forget, exaggerate, B.S., give up (balk), misinterpret, etc.

So, questions need to anticipate the struggles of studying human subjects.

Survey instruments have to be careful to produce accurate results and clear answers without **priming** respondents.

Descriptive vs. Inferential Statistics

Descriptive Statistics:

- Describes characteristics, properties, and general patterns in the data
- Relies on basic visual inspection and simple calculations identify trends in data that is already known (sample)
- Mean, median, mode, range, distributions, etc.

Inferential Statistics:

- Makes inferences (i.e., draws conclusions) about relationships in the population from relationships in the sample
- Uses specific techniques/tests to compare and *predict* relationships in data that is unknown (population)
- Relies on complex calculations and probabilities (e.g., p-values)
- Engages in systematic hypothesis-testing for statistical significance

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A little math doesn't imply a quantitative approach:

- Qualitative analysis can involve interpreting descriptive statistics and trends in charts/graphs.
- Formal modeling (game theory) uses complex mathematical equations.
- Quantitative analysis relies on specific tests of inferential statistics (e.g., chi-square, t-test, ANOVA, correlation, OLS regression, Logit, etc.).

Null Hypotheses

The **null hypothesis** is the default—that our hypothesis is *wrong*.

- It has to be mutually exclusive with the hypothesis of interest.
- Sometimes hypotheses are complex, though.
- The most basic form of a null hypothesis is **not** the *opposite* relationship, but the *lack* of a relationship.

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H₀: There is no relationship between Biggie's wealth and the number of problems he faces.

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Instead, we care about the probability of hypotheses being supported. There is always “error” (i.e., uncertainty). Humans are complicated.

0.05 is the threshold

The **p-value** is an indicator of of statistical *significance*

- How likely is it to get this result if the *null* hypothesis is true?
- Significance is often denoted using symbols (*)
- When $p < 0.05$: statistically significant (reject H_0 in support of H_1)
- When $p > 0.05$: *not* statistically significant (fail to reject H_0)
- **Note:** we say “not significant” rather than “insignificant”

Types of Errors

		The Truth (Based on Entire Population)	
		Nothing Is There (H_0 Is True)	Something Is There (H_0 Is False)
Your Conclusion (Based on Your Sample)	I Don't See Anything (Nonsignificant)	Right!	Wrong (Type II Error)
	I See Something (Significant)	Wrong (Type I Error)	Right!

Type I Error (false positive): cancer diagnosis when there is none

Type II Error (false negative): letting spam through the filter